SafeGuard AI: Fall and Drowning Detection

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Abstract

This project is a program that watches a camera or a recorded video and raises an alert when a person falls down in a room, or when a swimmer in a pool looks like they are in trouble. It finds people with a YOLO11 model, follows each person from frame to frame, and then applies rules about posture, speed and stillness to decide whether something is wrong. A person who goes down and then disappears from view is also treated as a fall. The alert is shown on screen, rings through the computer's speakers, and can be sent to a phone as a Telegram message with a photo. We built a Streamlit web page for demonstrations and a separate always-on service for a home, which restarts itself, reports a camera going offline, and sends a check-in message every morning. The fall detection worked in a live test with a phone camera. The pool detection has only been checked on made-up data and one short clip, and we say this clearly in the limitations.

1.Introduction

Many older people live alone or spend most of the day alone while their family is at work or in another city. If one of them falls and cannot get up, it can be a long time before anyone finds out. Drowning has a similar problem. A person who is drowning usually does not shout or wave. They often stay upright in the water, make little progress, and go under quietly, so a lifeguard looking at a busy pool can miss it.

Cameras are already cheap and common in homes and pools, but a camera only records.
Someone still has to be looking at it. The idea of this project is to let a computer do the looking and only call a human when something seems wrong.

2. Objectives

1. Detect a sudden fall of a person on land and raise an alert.
2. Detect signs of a swimmer in difficulty in a pool and raise an alert.
3. Work with a live camera link (including a phone used as a camera) and with an uploaded video, so the result can be shown to other people.
4. Send the alert to a phone with a photo, and sound a siren.
5. Keep the design simple enough to run on an ordinary laptop, and to move later to a CCTV camera.

3. How the system works

The program is one pipeline. A frame comes in from the camera, the people in it are found and tracked, the detection logic looks at how each person is moving, and if it decides there is an emergency the alert is sent.

3.1 Reading the video

Frames are read in a separate thread that always keeps only the newest frame. This matters for live camera links, because if the program is slower than the camera the old frames would pile up and the picture would drift further and further behind real time. We analyse about ten frames per second, which is enough for a fall and keeps the load low.

3.2 Finding and tracking people

People are found with a YOLO11 model from the Ultralytics library. For falls we use the pose version, which also gives body points such as shoulders and hips. For the pool we use the plain detection version. A tracker called ByteTrack gives each person an ID number and keeps it from frame to frame, so the program can follow one person over several seconds.

3.3 Fall detection

A fall is more than a person lying down, because people also sit, bend, crouch and do yoga. So the program looks at a short story rather than a single picture.

For every tracked person it measures the angle of the body (shoulders to hips) compared with straight up, the shape of the box around the person, how fast the hips are moving downward, and whether the person is moving at all. Stillness is measured from the movement of the box and from how many pixels change inside it. We first tried using the movement of the body points, but those jump around a lot on a person lying on the floor and the person never looked still.

Each person is in one of three states:

- **OK.** Nothing unusual.
- **Checking.** The person was upright a moment ago and is now down.
- **Fall.** The person has stayed down and mostly still for a few seconds.

If the person gets back up during the checking state, the check is cancelled. This is what stops most false alarms from sitting down or picking something up. There is also a second path. If someone is found lying still for about ten seconds without the program having seen the drop, it also raises an alarm, because the tracker may have started after the fall.

One problem showed up during testing. When a person falls quickly the picture is blurred and the tracker sometimes drops them and gives them a new ID a moment later. The new ID knew nothing about the person standing before, so the fall was never recognised. We fixed this by letting a new ID take over the history of a recently lost person who appears in nearly the same place.

A person can also go down and drop out of the picture, for example behind a sofa or past the bottom of the frame. The program remembers the last thing it saw. If the person was in the checking state when they vanished, or had just dropped suddenly in the middle of the frame, and they stay gone for about five seconds, it raises the alarm. A person who walks out through the edge of the frame, or steps behind a door while standing, does not trigger it. When an uploaded video ends right after someone goes down and disappears, the program reports it at the end instead of waiting.

The size of the YOLO model can be chosen. The small and medium pose models are more accurate than the nano one but slower, so on a laptop without a graphics card the medium model analyses fewer frames per second.

3.4 Drowning detection

For the pool the program looks at three signals for each swimmer and turns each into a risk value.
The highest one counts.

- The swimmer is upright and is not getting anywhere. A person who is drowning often stays vertical and moves very little across the pool.
- The swimmer is not moving at all, which suggests floating face down or sinking.
- The swimmer disappears in the middle of the pool for too long, which suggests being under water. If the person disappears next to the edge of the pool the program assumes they climbed out.

When the risk passes about 60 percent a warning is raised, and at 100 percent an alarm. If the tracker gives a returning swimmer a new ID, the new ID inherits the old timers, so the clock keeps running. A polygon drawn around the water ignores people walking on the deck.

3.5 Alerts

An alarm always shows a red banner with a snapshot. The siren sound is played through the speakers of the computer running the program, because web browsers often refuse to play sound that starts on its own. The length of the alarm can be set from 3 to 100 seconds. Optionally the alarm also sends a Telegram message with the photo, and it can switch a real siren connected to a Raspberry Pi and relay. Alerts of the same type from the same camera are held back for a minute so one event does not send a flood of messages. On the demo page, Telegram and the hardware siren are off by default, so a demonstration never sets off a real siren by accident.

3.6 The always-on home version

The demo page only works while its browser tab is open, so it is not suitable for watching a home all day. For that we made a second program that runs without a browser. It runs on a computer that stays on in the same house as the camera, starts the cameras by itself, and sends only the small alert over the internet, so the family gets the photo on their phone wherever they are.

A supervisor part checks the cameras every thirty seconds. If a camera program crashes, it is restarted, with longer and longer waits if it keeps failing. If a camera sends no video for ninety seconds, a Telegram message says the camera is offline and that nothing is being watched, and another message says when it is back. The program also sends a message when it starts, for example after a power cut, and a check-in message every morning. If the morning message does not arrive, the family knows the computer is off or has no internet. A camera stopped on purpose from the dashboard is not restarted, and if no camera link has been set yet the program still starts and says so.

4.Implementation

The project is written in Python. The main parts are listed below.

Part	Tool used
Detection and tracking	Ultralytics YOLO11 (pose and detection), ByteTrack
Image handling	OpenCV, NumPy
Optional learned fall check	PyTorch (a small LSTM)
Demo web page	Streamlit
Always-on service and API	FastAPI with uvicorn, and an SQLite event log
Phone alerts	Telegram Bot API
Siren	The computer's speakers, or a Raspberry Pi with a relay controlled over HTTP
Packaging	Docker and docker-compose

There are two front ends. The Streamlit page has an upload tab and a live link tab, and it is the one used for demonstrations. The FastAPI service has no page that needs to stay open, so it is the one meant for a home that needs watching all day. It is started with uvicorn, and a small Windows batch file starts it again if it ever stops. All detection settings are in one config.yaml file, and secrets such as the Telegram token and the camera link are kept in a separate .env file. The YOLO model files are not part of the project. They download by themselves on first use, and a short script downloads them again if they are deleted.

5.Testing and results

5.1 Logic tests

We wrote twenty-three small automated tests that feed made-up positions and fake cameras into the code. For falls, they check that a real fall gives one alarm, that a fall still gives an alarm when the tracker changes the person's ID in the middle, that a person found lying for ten seconds gives an alarm, and that a person who goes down and disappears from view gives an alarm. They also check the cases that should not alarm: sitting down, bending over and getting back up, walking, walking out of the frame through its edge, and stepping out of view while standing. For the pool, they check that an upright swimmer who stays in one place first gets a warning and then an alarm, that a normal swimmer gives nothing, that a swimmer who vanishes in the middle of the pool gives an alarm, and that a swimmer who leaves at the edge gives nothing. For the always-on part, they check that an offline camera and its recovery are each reported once, that a crashed camera is restarted with waiting between attempts, that a camera stopped on purpose is left alone, that the morning message is sent once a day, and that a camera with no link is ignored. All twenty-three pass. These tests show the rules do what we intended. They do not say how well the system works on real video.

5.2 Live test with a phone camera

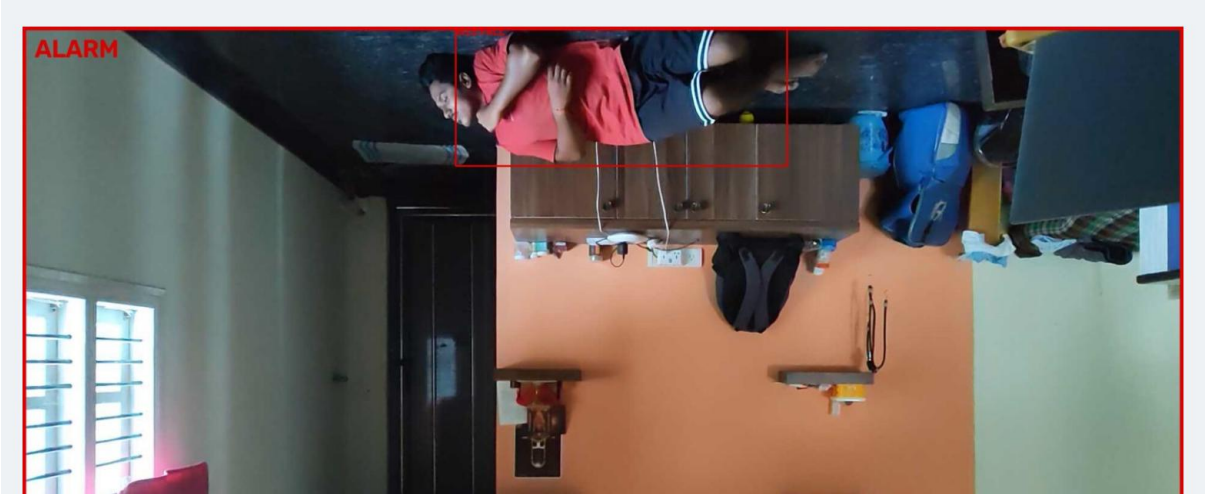
We used a phone running the IP Webcam app as the camera and the Streamlit page on a laptop. The first attempt failed because the link was missing /video at the end. After that was corrected, the live stream was read, the detection ran, and the alarm and siren sound worked.

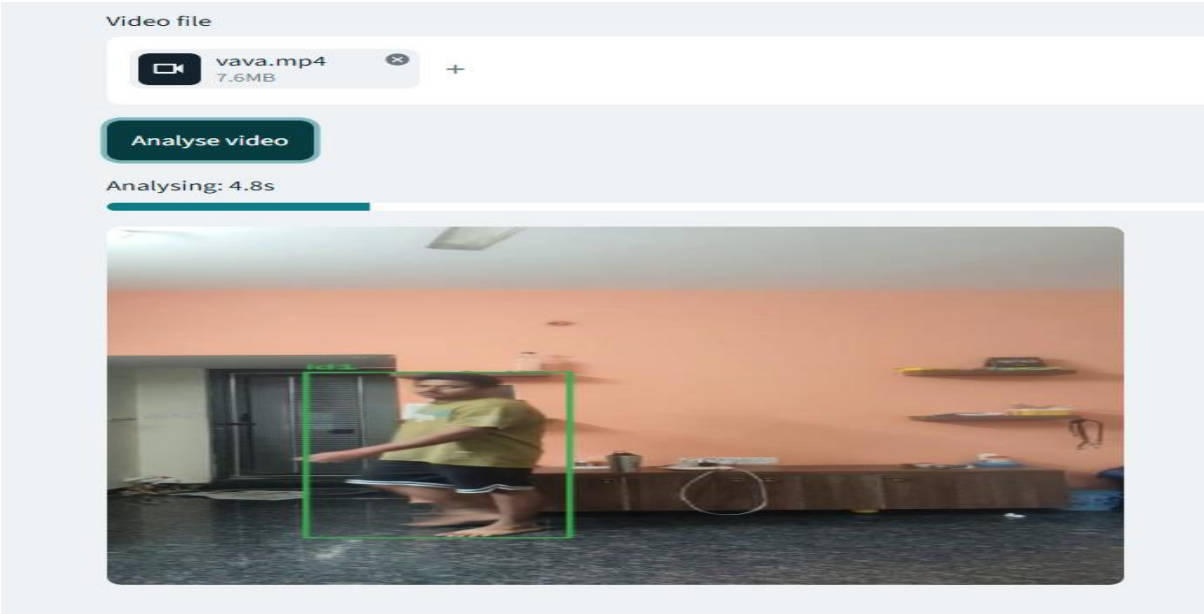
5.3 Recorded video

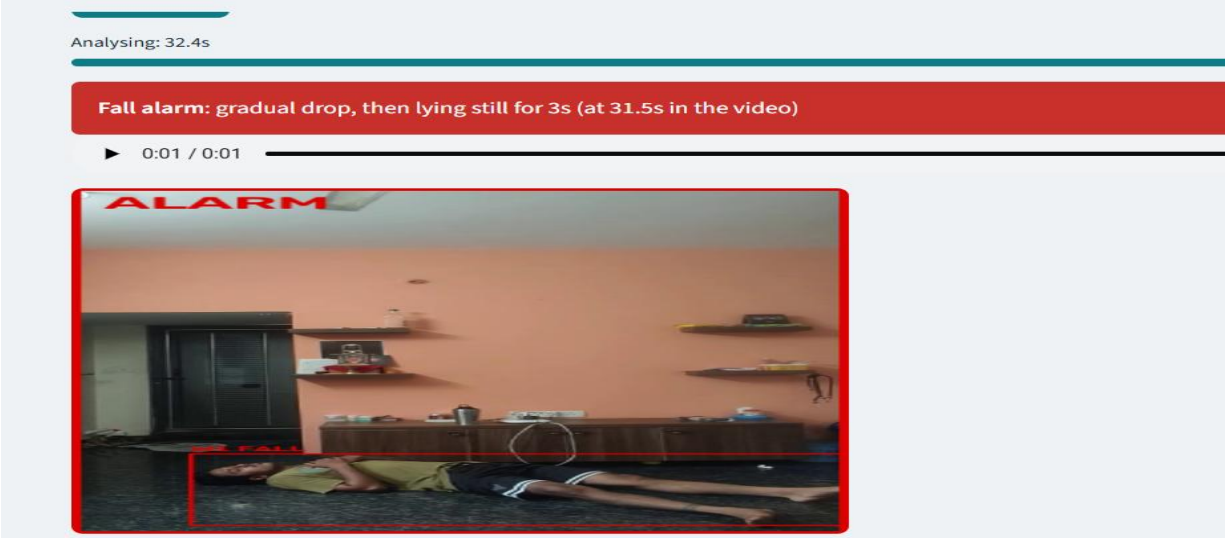
We recorded two clips of a person walking and then falling. In the first clip the person drops onto hands and knees toward the camera, partly out of frame, and the clip ends only about a second after the fall. No alarm was expected to fire there, because the program waits several seconds for the person to stay still, and it did not. In the second clip the person falls and lies flat for about twelve seconds, which should count as a fall, yet at first no alarm was raised. Reading the code again we found likely reasons: the stillness measure was based on jumpy body points, the tracker could change the ID during the fall, and the lying person was small in the frame at the default resolution. We changed the stillness measure, added the ID hand-over described above, raised the default detection size and lowered the default confidence. We have not yet repeated the second clip with the new settings, so we do not report a result for it here.

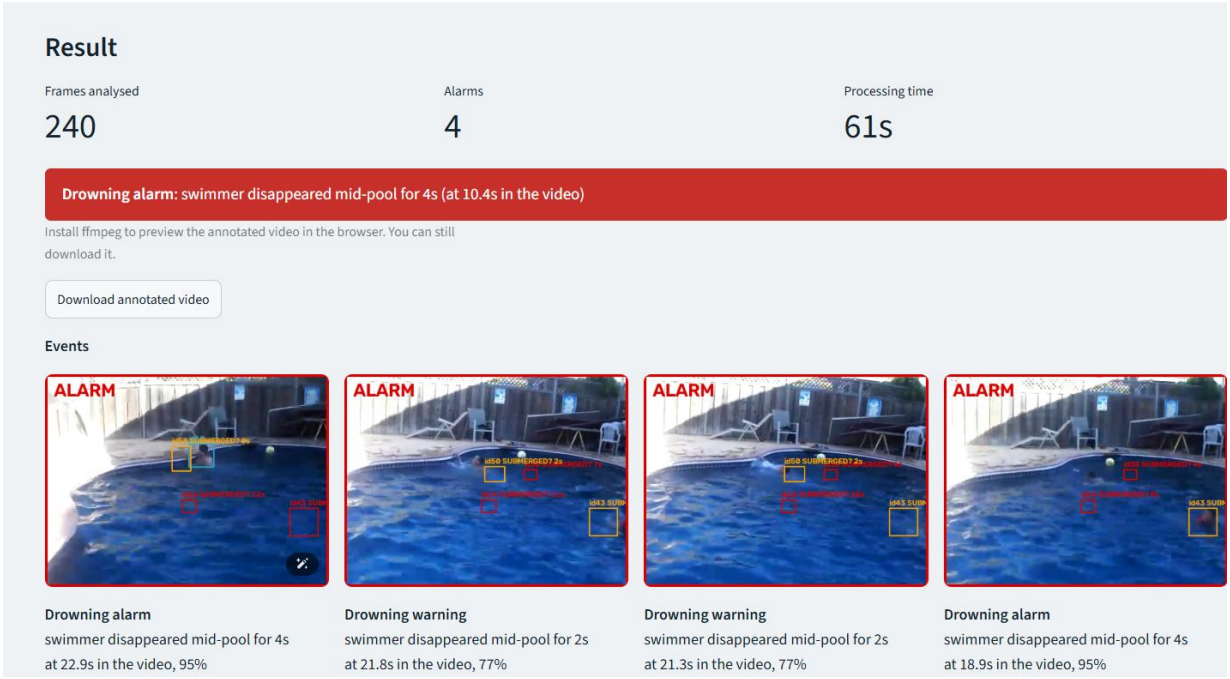
We also ran the pool mode on one short clip of a pool. The program drew its boxes and labels, and a "submerged" label appeared at one point, but we have not counted the results, so we do not claim any accuracy for the pool.

Results

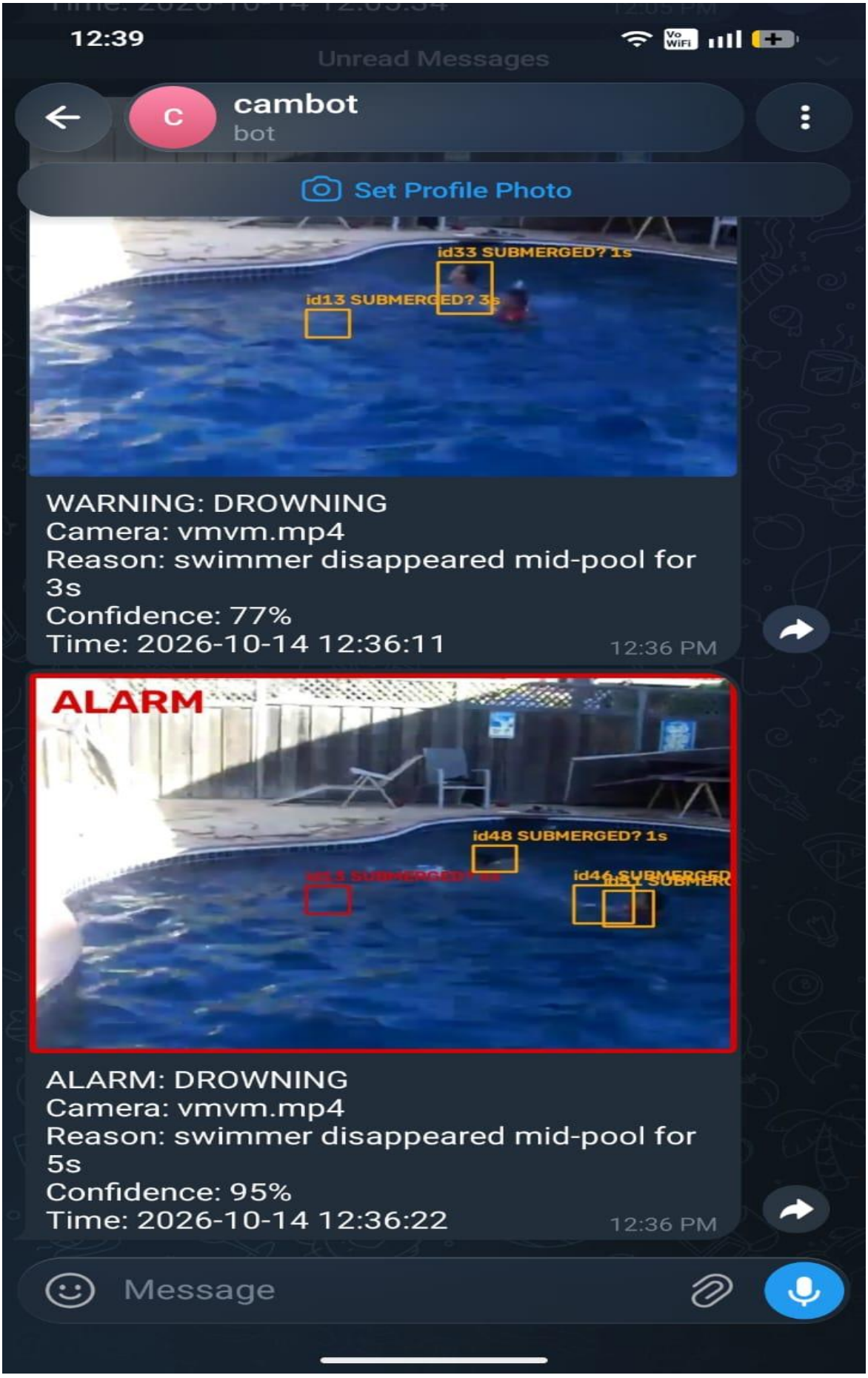

ALARM


Video file
vava.mp4
7.6MB
Analyse video
Analysing: 4.8s


Analysing: 32.4s
Fall alarm: gradual drop, then lying still for 3s (at 31.5s in the video)
0:01 / 0:01
ALARM


Result
Frames analysed
240
Alarms
4
Processing time
61s
Drowning alarm: swimmer disappeared mid-pool for 4s (at 10.4s in the video)
Install ffmpeg to preview the annotated video in the browser. You can still download it.
Download annotated video
Events
ALARM
ALARM
ALARM
ALARM
Drowning alarm
swimmer disappeared mid-pool for 4s at 22.9s in the video, 95%
Drowning warning
swimmer disappeared mid-pool for 2s at 21.8s in the video, 77%
Drowning warning
swimmer disappeared mid-pool for 2s at 21.3s in the video, 77%
Drowning alarm
swimmer disappeared mid-pool for 4s at 18.9s in the video, 95%


ALARM
ALARM: FALL
Camera: CCTV 1
Reason: gradual drop, then lying still for 3s
Confidence: 75%
Time: 2026-10-14 12:05:34
12:05 PM


12:39
Unread Messages
cambot
bot
Set Profile Photo
id33 SUBMERGED? 1s
id13 SUBMERGED? 3s
WARNING: DROWNING
Camera: vmvm.mp4
Reason: swimmer disappeared mid-pool for 3s
Confidence: 77%
Time: 2026-10-14 12:36:11
12:36 PM
ALARM
id48 SUBMERGED? 1s
id46 SUBMERGED
ALARM: DROWNING
Camera: vmvm.mp4
Reason: swimmer disappeared mid-pool for 5s
Confidence: 95%
Time: 2026-10-14 12:36:22
12:36 PM
Message

6. Limitations

- The system can miss a fall when furniture hides the person, when the room is dark, or when the person falls straight toward the camera so that the body is shortened in the picture.
- It can raise a false alarm when someone lies down on purpose, for example on a sofa or doing exercise.
- A fall that happens completely out of sight, with no drop visible beforehand, cannot be caught. The rule for a person who disappears only works if the program saw the person go down or drop first, and a person who goes out of view while standing is not counted.
- The pool detection is a first version based on rules. It has only been checked on made-up data and one short clip, not on real pool footage. A generic person detector often misses swimmers who are partly under water, and glare makes it worse. A model trained on pictures of the actual pool would be needed before anyone could rely on it.
- The learned LSTM check is included but has not been trained, because we do not have enough labelled clips.
- The medium YOLO model is slow on a laptop without a graphics card, so the program analyses only a few frames per second. A smaller model is faster but less accurate.
- A live camera link only works if the computer running the program can reach the camera. A camera in another city needs the program to run next to it, or a private network tool such as Tailscale.
- If the power or the internet goes off, monitoring stops. The always-on version cannot report its own power cut, so the morning check-in message not arriving is the warning sign.
- A camera records people. They should know about it and agree to it, and cameras should not be placed in bedrooms or bathrooms.

7. Future work

- Record a larger set of fall and daily-activity clips, measure how many falls are caught and how many false alarms happen per hour, and train the LSTM check on them.
- Collect and label footage of a real pool and train a swimmer detector on it.
- Run the always-on version for several days in a real home, with the family's agreement, and count the false alarms.
- Install the always-on version as a proper Windows service, so it starts even before anyone logs in.
- Add ignore zones, so a sofa or bed does not cause false alarms.
- Support several cameras on one computer, and run the model on a graphics card or a small edge device.

Conclusion

The project shows that a fall can be spotted from ordinary camera video using a pose model plus a few simple rules about posture, stillness and a person disappearing, and that the alert can reach a phone within seconds. The fall part worked in a live test and passed its logic tests. Testing on recorded clips showed real weaknesses, and we changed the program in response. We also turned the demo into a version that can run at home all day and report its own problems. The pool part is still an early version and needs real footage before it can be judged. The most useful next step is more recorded test clips, with honest counts of what was caught, what was missed and what raised a false alarm.